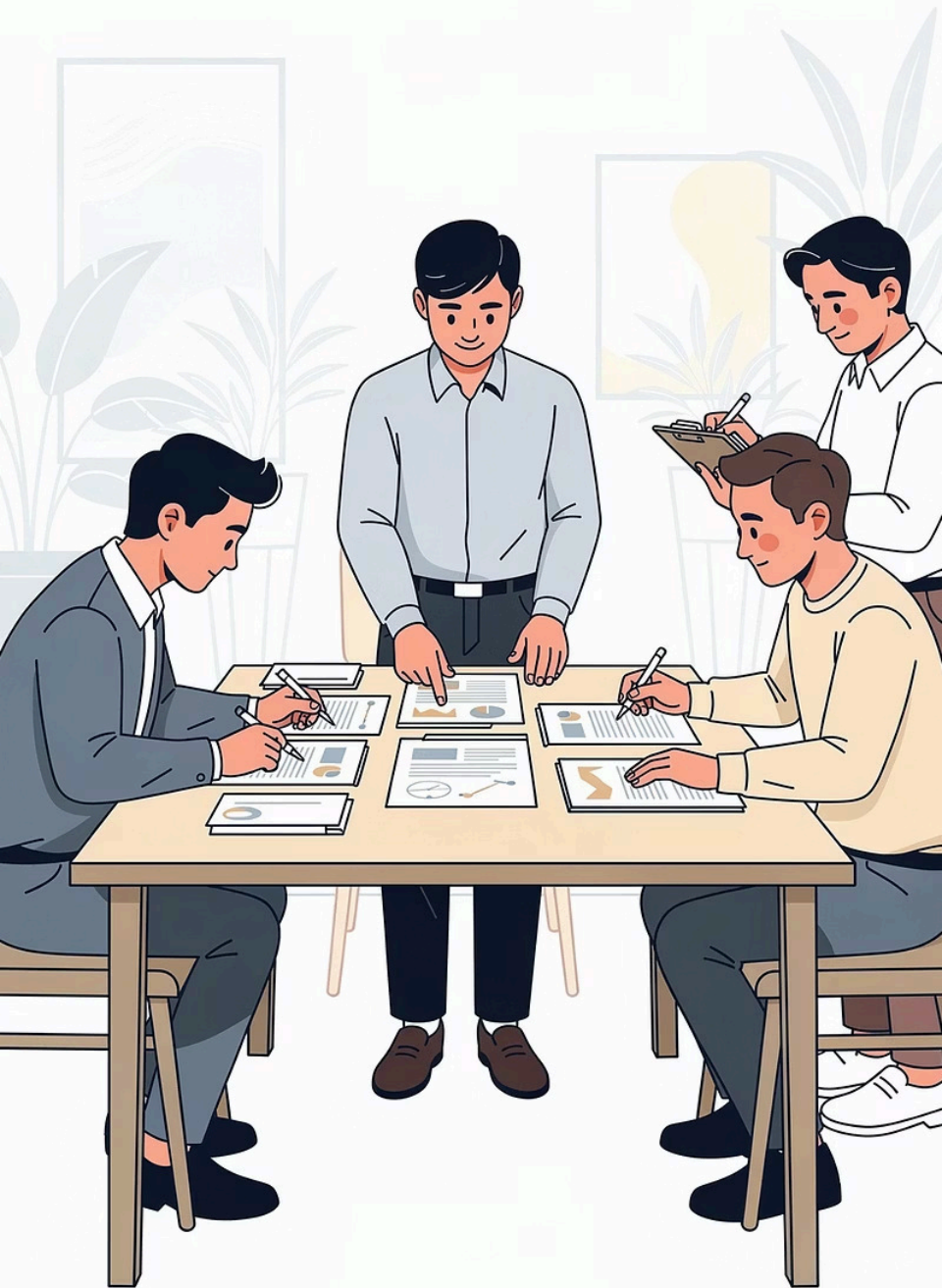




## Session 21: AI Safety, Misuse & Governance

Last session, you learned that AI can be unfair. Today, the bigger conversation: AI can also be **misused** - sometimes to seriously hurt people. But here's the good news: rules are being made to keep AI safe. And **YOU have a role to play.**

# Objectives

By the end of this session, you'll be able to:

01

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## Mistakes vs. Misuse

Tell the difference between AI misuse and AI mistakes

03

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## Privacy & Data

Understand AI privacy and why your data matters

05

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## Spot Deepfakes

Identify 3 telltale signs of a deepfake

02

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## Deepfakes

Explain what a deepfake is - and why it's dangerous

04

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## AI Governance

Know what AI governance means - rules to keep AI safe

06

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## Take Action

Know exactly what to do when you suspect an AI scam

# Recap & Homework

Welcome back! Let's see your **Bias Detective** work.

What 2 examples of bias did you find?

Who was affected? How might it be fixed?

What are the 3 sources of AI bias?

**Data · Algorithm · Human**

Name one real-world case

Gender Shades / Amazon Hiring / COMPAS



# Warm-Up

## Mistakes vs. On Purpose



### The Accident

Imagine a friend **accidentally spills** your juice. They didn't mean to. It just happened.

This is a **MISTAKE**.



### The Choice

Now imagine someone **takes your juice and pours it out on purpose** to mess with you.

This is **MISUSE**.

① Same result (no juice!) - but very different situations. **AI has both kinds of problems too.**

# AI Mistakes vs. AI Misuse

Two very different kinds of problems with AI:

## AI MISTAKES

AI gets something wrong **by accident**

- Hallucinations, bias, errors
- No one intended harm
- Solution: better data, better testing

## AI MISUSE

People **USE AI to do bad things on purpose**

- Scams, fake videos, stealing
- A human chose to cause harm
- Solution: rules and laws

 Mistakes happen. Misuse is when people **choose** to use AI for harm. Today is mostly about misuse - and what we do about it.



# Concept #1 - Deepfakes 🤖😬

**Deepfake** = a fake video, image, or audio made by AI that looks **REAL**.



## Fake Video

A video of a celebrity saying something they **never** said



## Fake Audio

An audio clip of your mom's voice - but **it's not really her**



## Fake Photo

A photo of you in a place **you've never been**



## Fake Live Stream

A "live stream" of a famous person - **totally fake**



The scary part: Modern deepfakes are really hard to spot. **Even experts get fooled.**

# How Are Deepfakes Made?

Remember Voice Cloning? That's one type of deepfake. Here's the recipe:



## 1. Collect Data

Photos, videos, or audio of a real person



## 2. Train AI

Copy their face, voice, or look



## 3. Generate

New content - putting them in scenes that never happened

It's basically **Generative AI** - used the wrong way.

# Real Story - The Voice Deepfake Scam

In 2019, an employee at a UK company received a phone call from someone who sounded **exactly** like his boss - urgently asking him to wire money to another company. He trusted the voice completely and sent it.

## \$240K

Lost to the scam

The company lost a huge amount of money  
in a single phone call

## 2019

Year it happened

Voice cloning has gotten WAY better since  
then

- ⊗ AI voices today can sound exactly like real people - even with the right emotion and pauses.





## A Different Real Story - The Ferrari Win (2024)

A scammer tried the SAME trick on a Ferrari executive. They used AI to fake the voice of Ferrari's CEO - asking the executive to help with a big secret deal.

But the executive noticed something off - the call was from a **different phone number** than usual. So he did something brilliant.

"What's the name of the book you recommended to me last week?"

The deepfake couldn't answer. The scammer hung up. The fraud was stopped. **Ferrari saved millions of dollars.** 💰

- ✓ The lesson: A simple "verify with something only the REAL person knows" works against even the best deepfakes.

**PAUSE → CHECK → ACT.**



# Other Ways AI Gets Misused

Deepfakes aren't the only AI misuse. Other ways bad guys use AI:

## Smarter Phishing

AI writes fake emails that look 100% real. *"Your Netflix is about to be cancelled - click here!"*

## Faster Hacking

AI helps criminals guess passwords or find weak spots in apps faster than any human.

## Disinformation at Scale

AI can pump out thousands of fake news articles in minutes, flooding social media.

## Harassment

Bad guys use AI to create mean fake content about real people.

 The lesson: AI gives **BAD humans superpowers too**. Which is why we need rules and safeguards.

# How to Stay Safe from Deepfakes

5 smart tricks to protect yourself:

## Call Back

If a voice asks for something weird → hang up and call that person at a **number you trust**.

## Use a Code Word

Pick a secret word with your family. If a "family member" calls in an emergency, **ask for the code**.

## Check the Source

Where did this video come from? Is it from a **trusted news site**?

## Look at the Details

Deepfake videos sometimes have weird eyes, blinking, hair, or audio that **doesn't quite match the mouth**.

## When in Doubt - Ask an Adult

Especially before sending money, photos, or personal info. **Smart kids = safe kids.** 💪

# Mini Activity 1 - Spot the Risk!

For each situation, what should you do?

| Situation                                                                 | What would YOU do? |
|---------------------------------------------------------------------------|--------------------|
| A voicemail "from Mom" asks for the credit card number                    | ?                  |
| A video shows your favorite YouTuber saying something mean                | ?                  |
| A "friend" texts asking to borrow money - and the voice note sounds weird | ?                  |
| You see a viral video of a celebrity announcing something shocking        | ?                  |
| Someone texts you a photo of YOU somewhere you've never been              | ?                  |

# Mini Activity 1 - Smart Answers

| Situation                   | Smart Move                                                                                                                      |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| "Mom" asks for credit card  |  Call mom back at a number you trust         |
| Mean video of YouTuber      |  Check official sources - could be fake      |
| "Friend" asks for money     |  Call them directly - could be voice cloning |
| Viral shocking video        |  Wait - see if real news confirms it         |
| Photo of you somewhere fake |  Tell a parent - could be edited             |

 Rule of thumb: **PAUSE → CHECK → THEN ACT**. Don't react fast. Real urgency is a trick deepfakes use.

# Concept #2 - Privacy & Your Data

Every time you use an app, **AI is learning about YOU.**

 **Location**

Where you go

 **Shopping**

What you buy

 **Viewing**

What you watch

 **Social**

Who you talk to

 **Gaming**

What games you play

 **Emotions**

What makes you happy or sad

 AI uses your data to make decisions about you - what ads to show, what news to feed you, even what prices to charge!

# Why Your Data Matters

3 reasons to care:

**Companies use your data to make money**

They sell it or use it to show you ads. Your attention is their product.

**Once data is out, it's hard to take back**

A photo you post stays on the internet forever - even if you delete it.

**AI gets BETTER at predicting you**

The more data it has, the more it knows. That can be helpful - but it can also feel creepy.

  **Your data = your story.** You should have a say in who gets it.

# How to Protect YOUR Privacy

Easy steps anyone can do:

## Guard your identity

Don't share your full name, address, school, or phone publicly online

## Check app permissions

Does this game REALLY need your location? Question every permission.

## Use strong passwords

Use DIFFERENT ones for different sites - never reuse passwords

## Don't click weird links

Even from "friends" - their account might be hacked

## Ask before posting photos of friends

Respect others' privacy the way you want yours respected

## Talk to a parent before downloading new apps

Small habits = big protection. 🛡️

# Concept #3 - AI Governance 📜

**Governance** = the RULES governments make to keep AI safe. Just like there are rules for driving cars, building houses, and selling food - there are rules being made for AI.

What AI CAN be used for

What AI is NOT allowed to do

When humans **MUST** stay in charge

What companies **MUST** tell us

- ⊗ Without rules, anyone could build any AI for any reason. **THAT would be dangerous.**



# Real Rules Being Made

Around the world, governments are writing AI rules.

## EU AI Act

Europe's big law about AI. Bans scary AI uses like predicting "future crimes." Forces companies to be honest about what AI they use.

## US AI Bill of Rights

Promises for how AI should be used. People have the right to know when AI is making decisions about them. People deserve to be safe from unfair AI.

## GDPR

Europe's data protection law. Companies **MUST** tell you what data they collect. You have the right to ask them to **DELETE** your data.

 These laws aren't perfect. But they're a start.

# The 3 Big Principles of AI Governance

Almost every AI law focuses on 3 things:

## 1 Transparency

Companies must **tell people** when AI is being used. No hiding it.

## 2 Accountability

If AI hurts someone, **someone must be responsible**. No passing the blame.

## 3 Human Oversight

Humans **MUST stay in charge** of big decisions. (Human-in-the-Loop! From Session 10!)

✓ Notice anything? **Transparency · Accountability · Human Oversight** = the 3 big rules. Easy to remember.

# Mini Activity 2 - Rule the AI World!

Imagine **YOU make the rules for AI**. What would you say YES or NO to?

| AI Use                                               | YES or NO? | Why? |
|------------------------------------------------------|------------|------|
| AI that watches kids on a playground for safety      | ?          |      |
| AI that decides who gets into a school               | ?          |      |
| AI that pretends to be a famous person to sell stuff | ?          |      |
| AI that helps doctors find diseases                  | ?          |      |
| AI that secretly listens to your conversations       | ?          |      |
| AI that grades your school essays                    | ?          |      |

Tell your mentor your reasons!

# Mini Activity 2 - Common Answers

Most people would say:

| AI Use                     | Common Answer                                   |
|----------------------------|-------------------------------------------------|
| Watch kids for safety      | 🟡 Maybe - only if parents know and agree        |
| Decide school admissions   | 🟡 Only if a human reviews + students can appeal |
| Pretend to be a celebrity  | ❌ Without permission - that's identity theft    |
| Help doctors find diseases | ✅ Yes (with doctor oversight)                   |
| Secretly listen            | ❌ Definitely no (privacy!)                      |
| Grade essays               | 🟡 Only if a teacher reviews                     |

✅ AI is OK when...

Humans stay in charge and people are informed

❌ AI is NOT OK when...

It's secret, replaces humans entirely, or steals identities

# Hands-On Activity - Deepfake Detective!

Time to put your deepfake-spotting skills to the test. I'm going to describe 3 mystery situations.

Your job: **Real or Fake? And how can you tell?**

Tells to watch for:

## Eyes & Blinking

Deepfakes sometimes blink weirdly or stare too much

## Mouth & Sync

Lips don't perfectly match the audio

## Hands & Details

AI struggles with hands, jewelry, glasses, teeth

## Voice Quality

Slight robotic tone, weird breathing, flat emotion

## Context

Does the source make sense? Why is this being shown to YOU?

# Detective Cases - Real or Fake?



## Case 1: The Presidential Video

A viral video shows a famous President saying something shocking. You see it on a random TikTok account with no source. His mouth moves slightly out of sync with the words.

**Your verdict: Real / Fake? Why?**



## Case 2: The Grandma Voice Message

You get a voice message from "Grandma" asking you to send a gift card right away because she's in trouble. Her voice sounds right but her words feel... off. She'd usually say "honey" but she said "kid."

**Your verdict: Real / Fake? Why?**



## Case 3: The Athlete Photo

A photo shows your favorite athlete holding a product, smiling. Their fingers in the photo look like they have 6 fingers instead of 5. It's posted on the brand's official page.

**Your verdict: Real / Fake? Why?**

# Detective Cases - Answers 🏆

| Case                       | Verdict                                                                                       | Key Tell                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1. President viral video   |  Likely FAKE | No trusted source + mouth out of sync. Real news clips come from real news sites.                  |
| 2. "Grandma" voice message |  Likely FAKE | Voice cloning is real. The "off word choice" is a huge red flag. CALL HER BACK at her real number. |
| 3. Athlete photo           |  Likely FAKE | AI image generators famously mess up hands. Count the fingers.                                     |

🕒 Your superpower: You now know the tells. Most adults DON'T. **You're literally safer than most grown-ups when it comes to deepfakes.** 💪

# What We Learned in this Session

## Mistakes vs. Misuse

Same result, different intent - and very different solutions

## Deepfakes

Fake media made by AI; can be spotted with the right tells

## PAUSE → CHECK → THEN ACT

The universal scam-stopper rule - never react fast

## Privacy Matters

Your data tells your story - protect it

## Governance

Transparency · Accountability · Human Oversight

## You Spotted 3 Deepfakes

You're already safer than most adults 

👉 You're now **AI-aware AND AI-safe.** 



# Your Homework - Safety Conversation

This week, have a **SAFETY TALK** with your family.

## Pick ONE adult at home

Parent, older sibling, anyone - and tell them what you learned about deepfakes today

## Together, pick a family code word

To use in real emergencies - keep it secret!

## Share the rule

Teach them: **PAUSE → CHECK → THEN ACT**

Write down: who you talked to, your family code word (keep it secret!), and what surprised them most.

# Next Time - Module 8 Begins!

You've completed every concept. Now you do **real AI research**. Over the next 4 sessions, you'll create your own AI investigation - like a real researcher.

## Session 22

### **Discover & Define**

Pick your topic, ask a research question, find sources

## Session 23

### **Research & Build**

Dig deep, run your own AI experiments, gather findings

## Session 24

### **Analyze & Synthesize**

Apply everything you've learned, polish your work, build your slides

## Session 25

### **Present & Reflect**

Share your findings, defend your ideas, celebrate

